; use horizontally, with in units of . Mark key points. For

, enter ranges below. (Only the graph and Table are scored.)

Table

range

5.

A clear semicylinder, index , radius , is in air.
The figure is a normal section; is its axis intersection.
Light enters only through plane . A broad parallel
monochromatic beam, normal to the axis, covers at angle .
Some light passes through and exits the curved surface. Its
width on along is . Exclude rays reflected
one or more times inside the cylinder before exiting.

. As incidence angle varies over , find the minimum of ,
and its maximum

. In the figure plane, relate the arc subtended at by the exit beam to .
Also locate this arc at grazing incidence.

6.

General relativity bends light in a stellar gravity field. The ray is in a plane through the centre, bending slightly near it. Its closest point is periapsis. A line through periapsis and centre is its symmetry axis. In polar coordinates , take the centre as origin, symmetry axis as polar axis. The photon trajectory is

is the gravitational constant, stellar mass, light speed; is far below in magnitude. A star is light years away; a white dwarf lies halfway to Earth.
At Earth, rays past either side subtend . Find the dwarf mass (kg).
Given